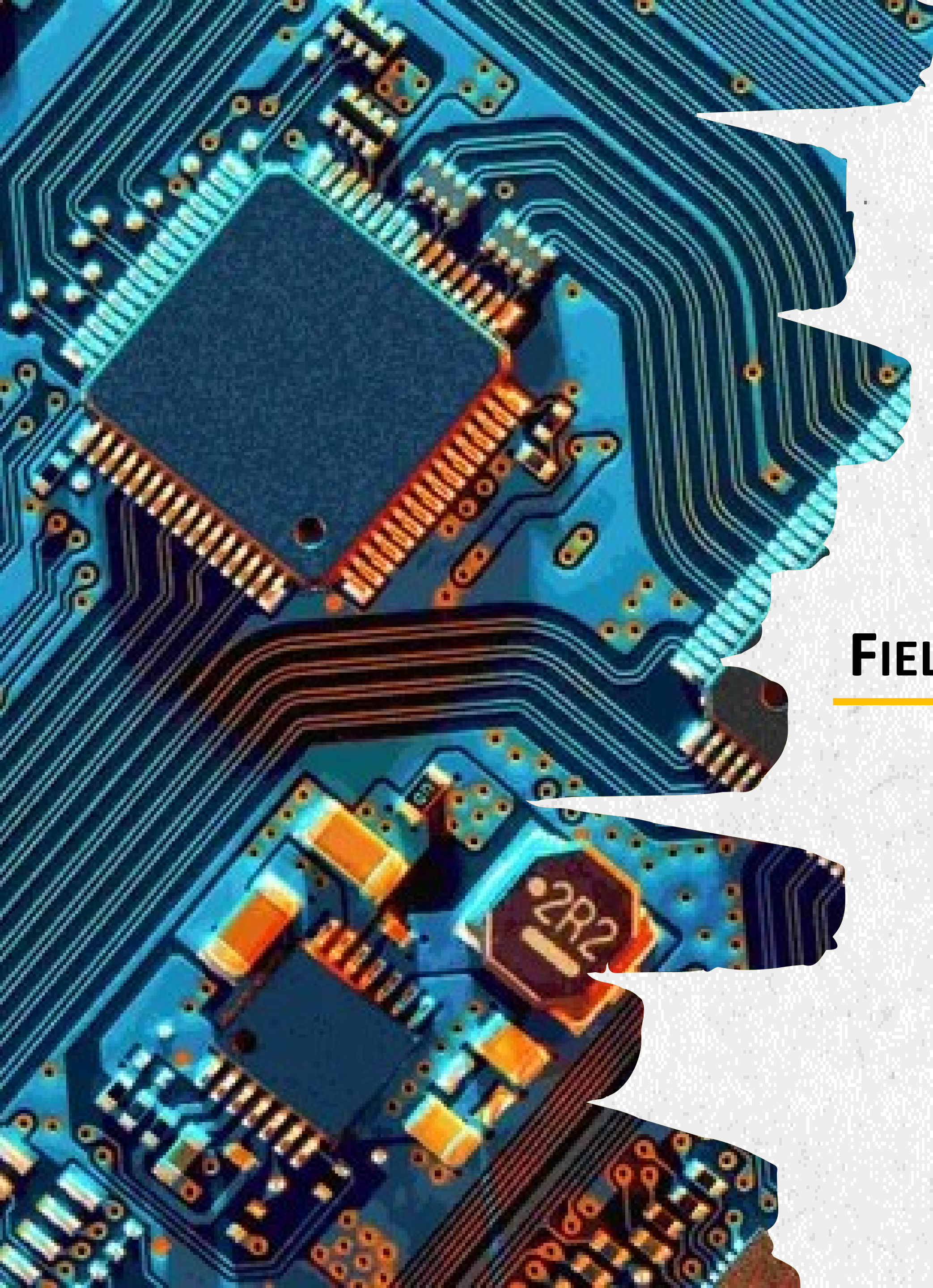
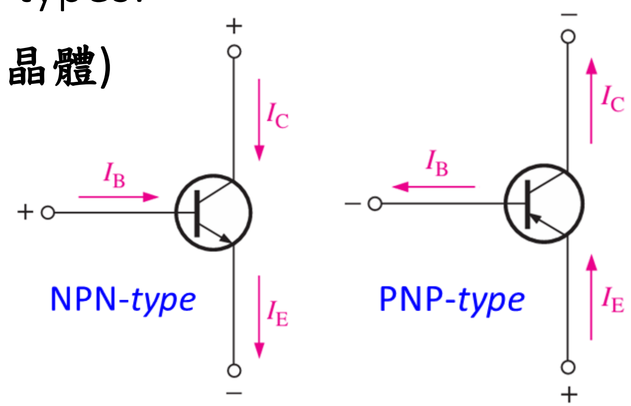


CHAPTER 8

FIELD-EFFECT TRANSISTORS (FETs)

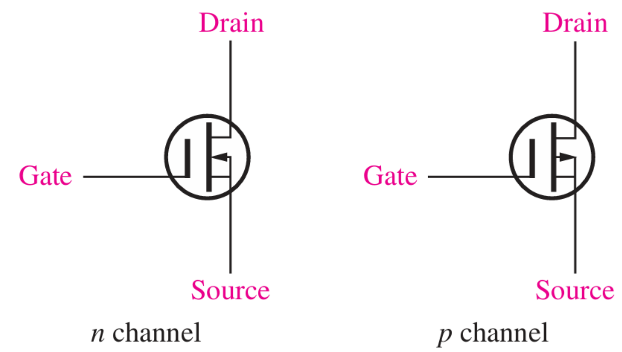
FIELD-EFFECT TRANSISTOR (FET)

- As introduced, transistors (電晶體) have two main types:
  1. Bipolar Junction Transistors (BJTs、雙載子連接電晶體)
     - 3 Pins – Base 基, Collector 集, and Emitter 射
     - 2 Types – NPN and PNP
     - Current (I_B) controls current (I_C)
     - *Applications*: basic switching and amplifier



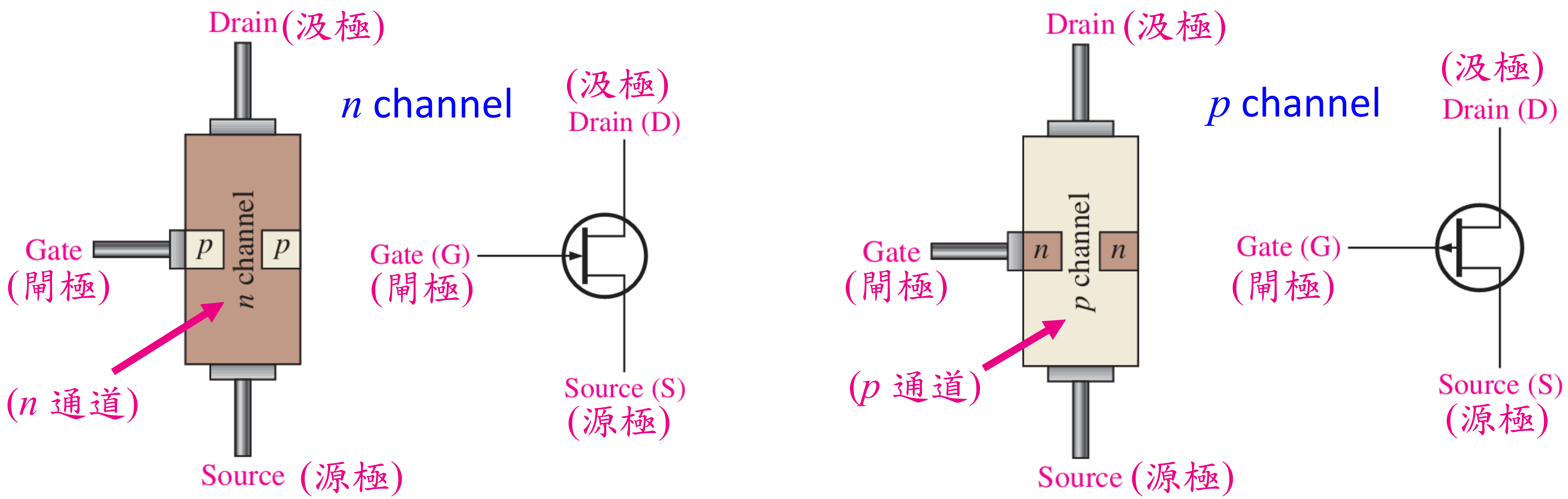
  2. Field-Effect Transistors (FETs、場效電晶體)
     - 3 Pins – Gate 閘, Drain 汲, and Source 源
     - 2 Types – n channel 通道and p channel 通道
     - Voltage (V_{GS}) controls current (I_D)
     - *Applications*: high-frequency high-voltage switching
     - *Most popular*: JFET and MOSFET

MOSFET



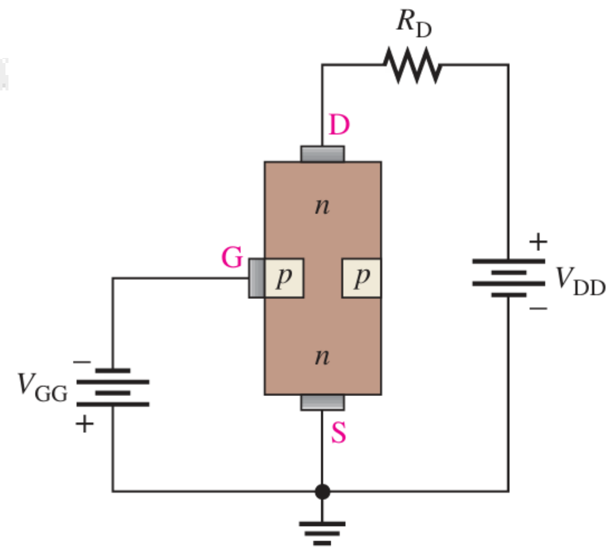
JUNCTION FIELD-EFFECT TRANSISTOR (JFET)

- The ***junction field-effect transistor*** (JFET、接面場效電晶體) is a type of FET that operates with a ***reverse-biased*** *pn junction to control current* in a channel.

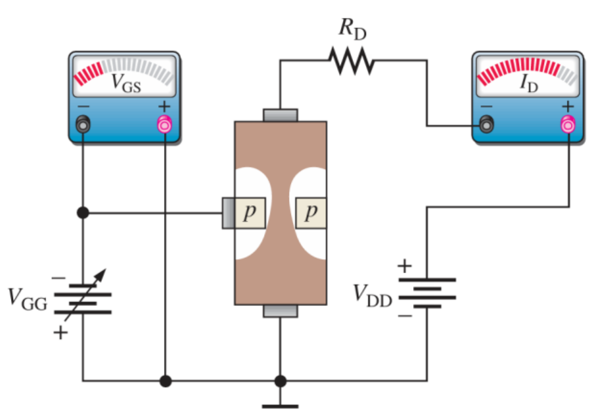


➢ Note that the *p*-(or *n*-) type region surrounds the *n* (or *p*) channel, like a belt.

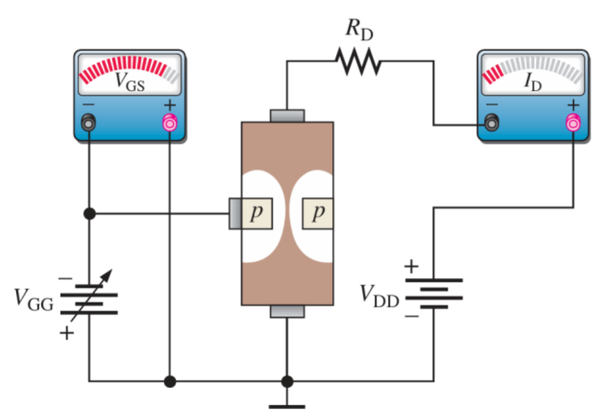
JFET BASIC OPERATIONS



- The ***JFET is always operated*** with the gate-source *pn* junction ***reverse-biased***.
- ***Reverse-biasing of the GS junction*** produces a **depletion region 空乏區** along the *pn* junction
  - Block the channel width
  - Increase the resistance of the channel



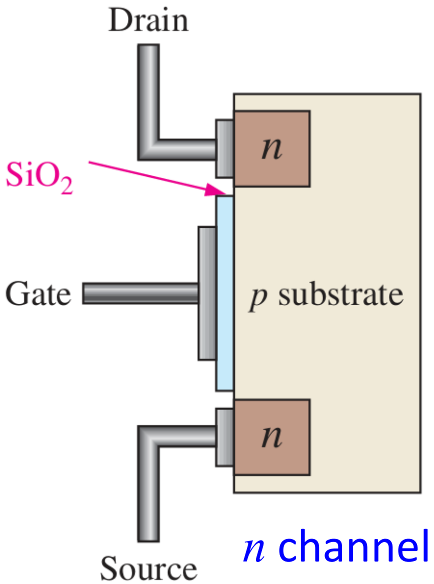
Less V_{GG}, Wider channel
Smaller resistance

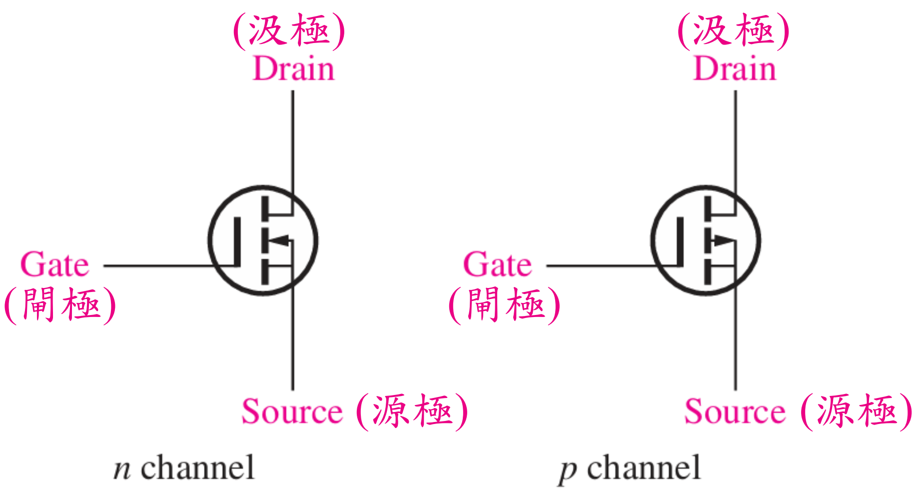


More V_{GG}, Narrower channel
Greater resistance

MOSFET

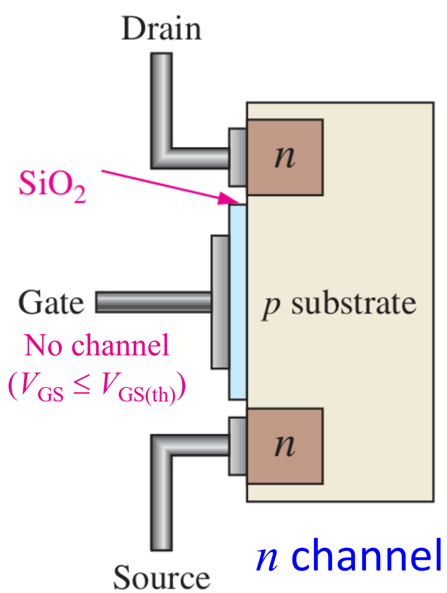
- The *metal oxide semiconductor field-effect transistor* (MOSFET、金屬氧化物半導體場效電晶體) is another category of field-effect transistor.
- The gate of the MOSFET is insulated from the channel by a silicon dioxide (SiO_2、二氧化矽) layer. SiO_2 is found in rock and sand, to make glasses.

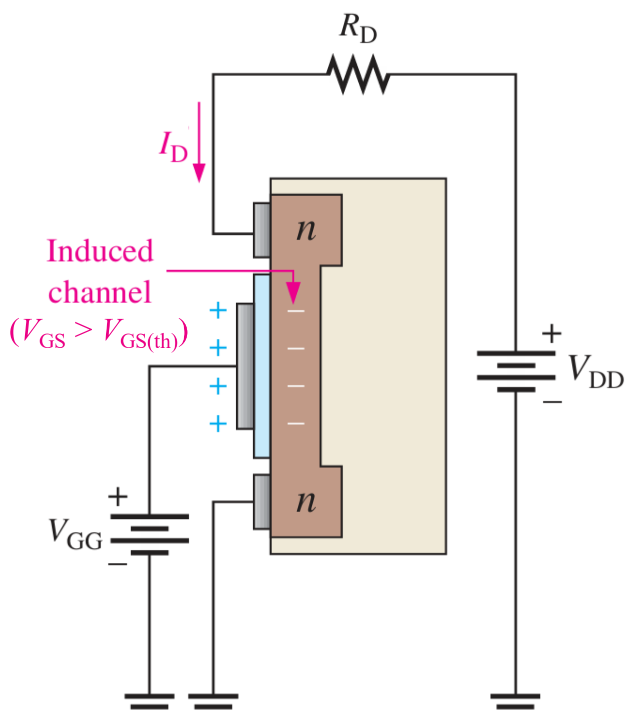


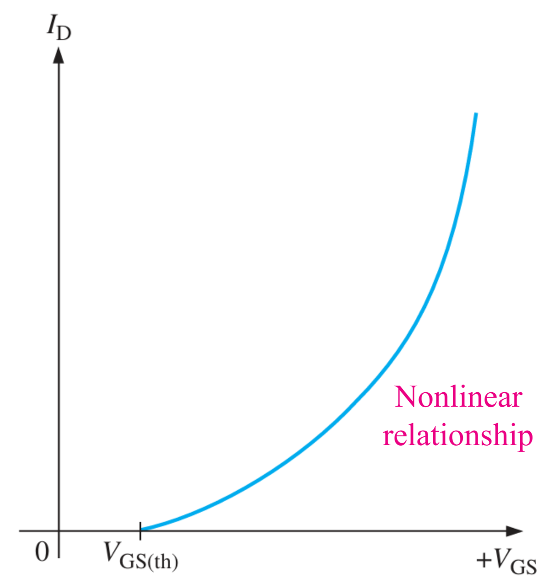


MOSFET

- A ***positive gate voltage above a threshold value*** ($V_{GS} > V_{GS(th)}$) induces a channel by creating a thin layer of negative charges in the substrate region.
- The conductivity of the channel is enhanced by increasing the gate-to-source voltage. For any gate voltage below the threshold value, there is no channel.

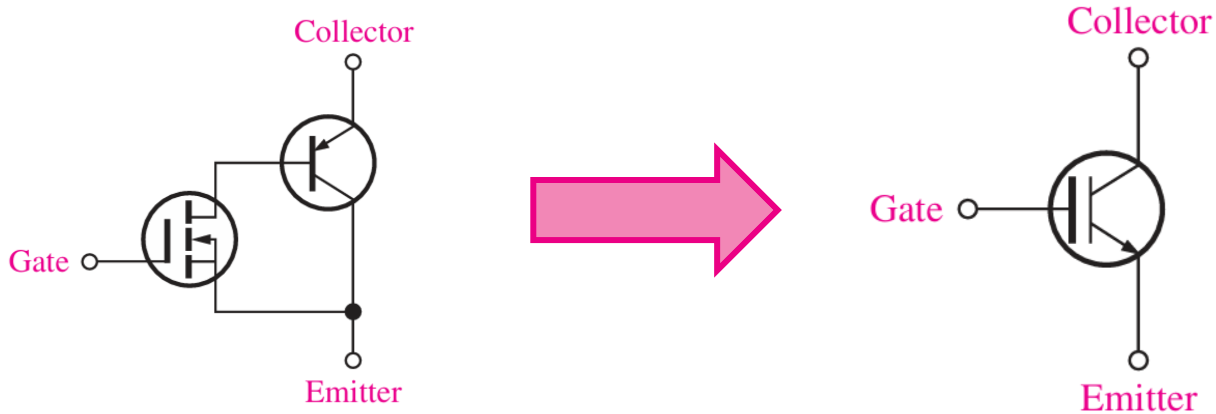






IGBT

- The ***insulated-gate bipolar transistor*** (IGBT、絕緣閘雙極電晶體) combines features from both the *MOSFET and the BJT* that make it useful in *high-voltage and high-current switching applications*.



- The IGBT is *voltage-controlled* like a MOSFET, but has the output conduction characteristic of a BJT, which is *capable of higher currents* than MOSFETs.

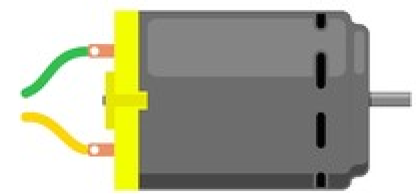
COMPARISON

FEATURES	IGBT	MOSFET	BJT
Type of input drive	Voltage	Voltage	Current
Input resistance	High	High	Low
Operating frequency	Medium	High	Low
Switching speed	Medium	Fast (ns)	Slow (μs)
Saturation voltage	Low	High	Low

- Hence, nowadays, the IGBT has largely replaced the MOSFET and the BJT in many of these applications.

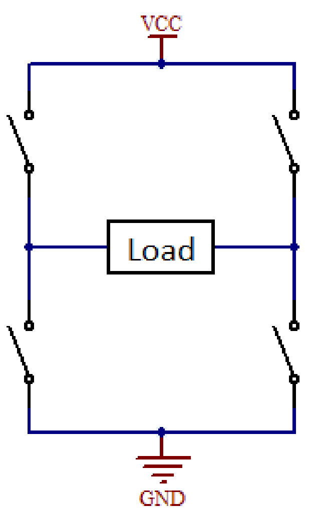
APPLICATIONS: MOTOR DRIVER

- A *controller or computer* works at **low voltage levels, 0V (LOW) and 5V (HIGH).**
- *DC motors* work in many different operating **voltage ranges (6V ~ 600V), and require sufficient current or power.**
- You can use BJT, MOSFET, or IGBT to control DC motors.

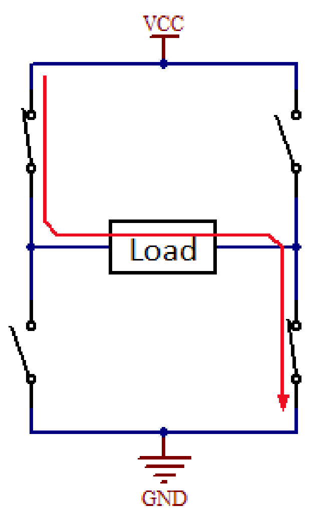


DC Motor

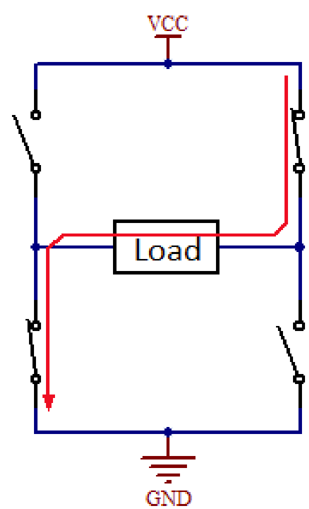
H bridge topology

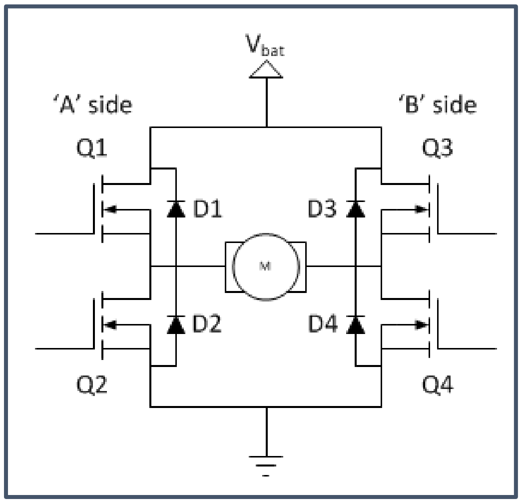


Connecting the load in one direction

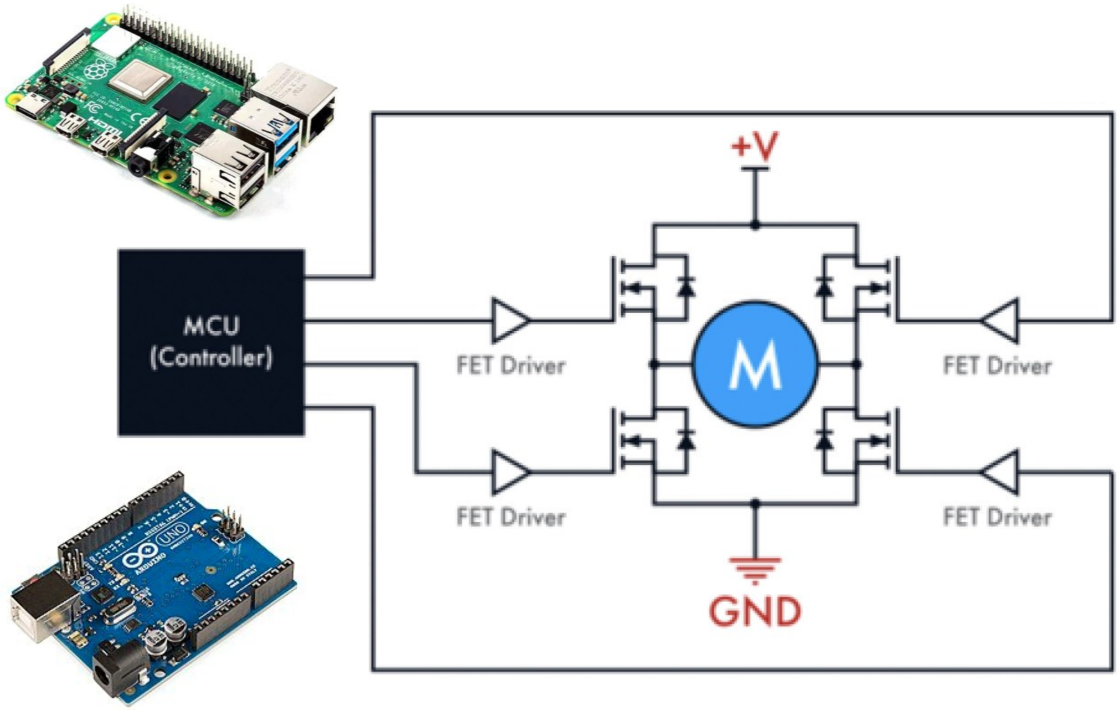


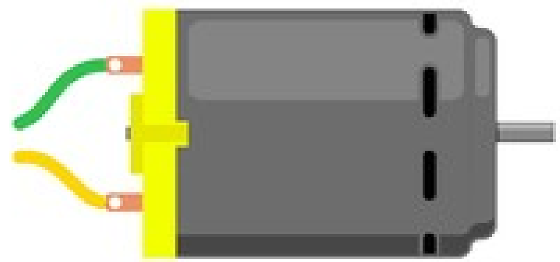
Connecting the load in the other direction





APPLICATIONS: MOTOR DRIVER





DC Motor

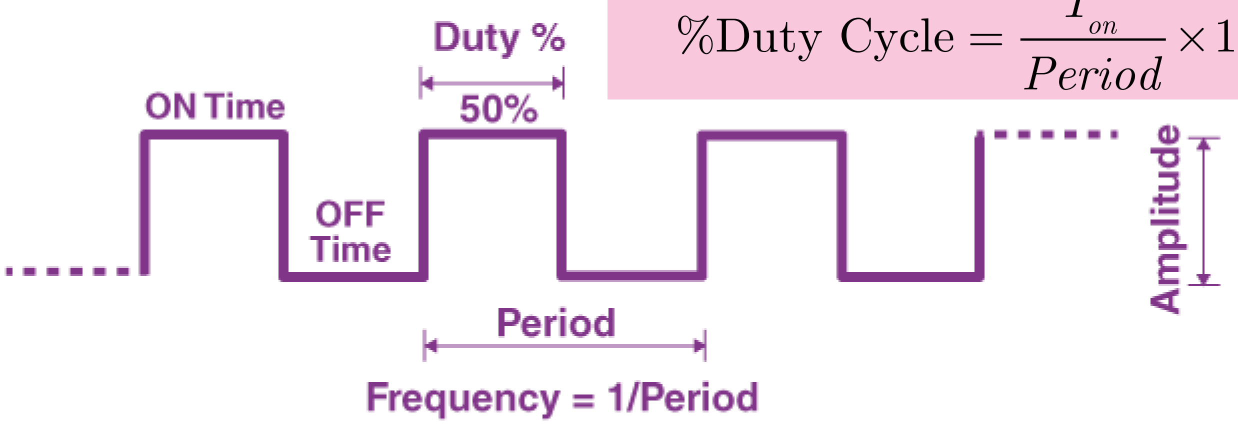
Microcontroller works in digital signals (Low ↔ High).

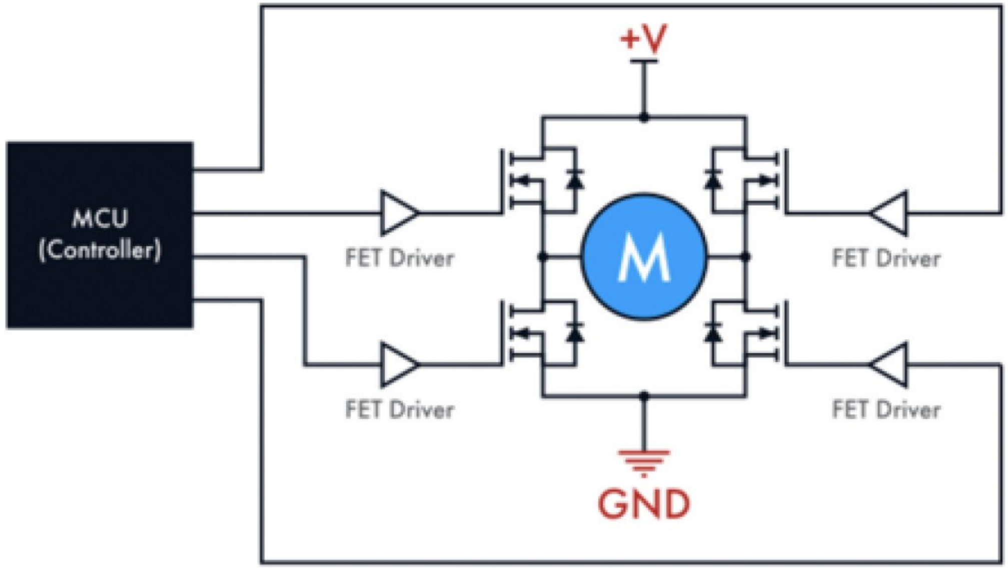
How do we make **analog control** of **the motor speed** ?

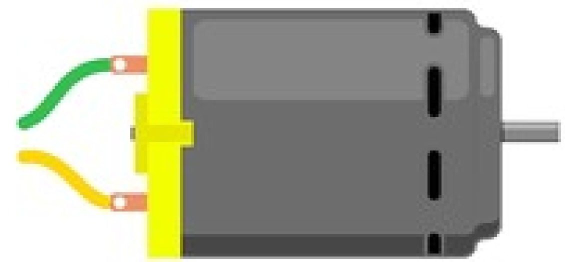
PULSE-WIDTH MODULATION (PWM)

- Pulse-width modulation (PWM, 脈波寬度調變) rapidly switches a signal On and Off by varying percentage of time the signal is "High" (% Duty Cycle) within a consistent frequency.
- This technique controls the average power or voltage delivered to a load.

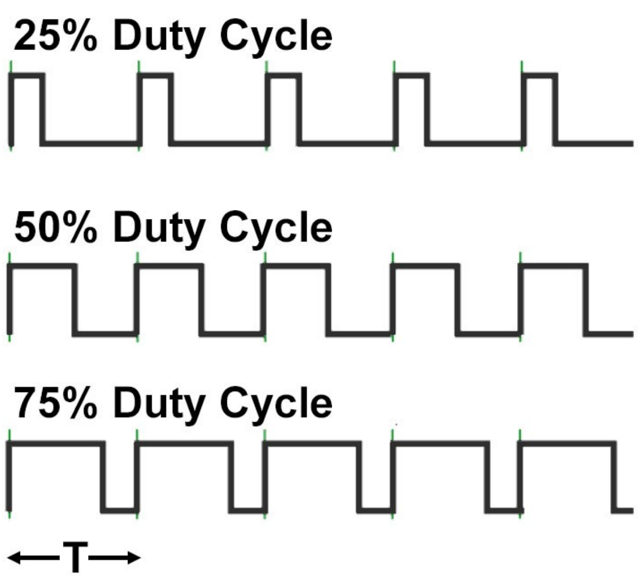
$$\%\text{Duty Cycle} = \frac{T_{on}}{Period} \times 100\%$$







DC Motor



25% of Rated Motor Speed

50% of Rated Motor Speed

75% of Rated Motor Speed

APPLICATIONS: INVERTERS

- Inverters (逆變器) are electronic devices to convert the DC voltage (直流電) to the AC voltage (交流電).